AI POLYGRAPH // LIE DETECTOR SIMULATOR

Complete Technical Architecture, Operational Scope, Capabilities & Forensic Specification

Project: AI Lie Detector Simulator	**Date:** October 2026
Core Engine: OpenCV Hardware Vision + Web Audio	**Status:** Production Ready & Calibrated
Platform: Python 3.12 / Flask / PyCharm IDE	**Classification:** Biometric Interrogation Simulation

1. What It Does (Core Capabilities & Features)

The system acts as an interactive biometric interrogation polygraph that captures and analyzes facial dynamics, gaze direction, and vocal tension in real-time to determine truthfulness versus deception.

- **Direct Hardware Webcam Capture:** Connects directly to hardware camera index 0 (with one-click switching to camera 1) bypassing browser WebRTC permission restrictions.
- **rPPG Real Heart Rate Extraction:** Contactless cardiovascular pulse (BPM) extraction from forehead skin color micro-fluctuations via Fast Fourier Transform (FFT).
- **Real-Time Live Speech-to-Text & Linguistic Analysis:** Transcribes the user's spoken words live (English/Hindi) and detects verbal evasion, emphatic protestations ('I swear', 'honestly', 'kasam se'), and cognitive hesitation fillers.
- **Real-Time Face & Eye Detection:** Employs dual Haar Cascade classifiers (Frontal Face & Dual Eye ROIs) processing 30 frames per second.
- **Eye Contact Stability Tracking (0–100%):** Continuously evaluates whether the subject is maintaining direct eye contact with the camera lens or looking away.
- **Gaze Aversion Detection:** Identifies sideways glances, downward head tilts, and rapid eye movements that correlate with cognitive deception or nervousness.
- **Adaptive Blink Dynamics & Frequency:** Tracks total blinks and calculates Blinks Per Minute (BPM), detecting both rapid anxiety blinking (>26 BPM) and conscious frozen stares.
- **Voice Pitch & Micro-Tremor (Jitter) Analysis:** Microphonic capture via Web Audio API, calculating fundamental pitch stability and vocal tension.
- **Multi-Channel Polygraph Oscillograph:** Real-time animated oscilloscope rendering 4 concurrent physiological channels: Cardio/Pulse ECG, Voice Tension, Eye Tension, and Deception Index.
- **5-Second Active Interrogation Scan:** Triggered via button or Spacebar, scanning the subject's response window with progressive tension audio and laser sweep.
- **Dramatic Holographic Verdict Screen:** Slams a high-impact, full-screen holographic verdict: **LIE DETECTED!** (Red sirens, alarm audio, forensic breakdown) or **TRUTH VERIFIED!** (Green glow, victory chime, credibility seal).

2. What It Does NOT Do (Operational Limits & Ethics)

To ensure responsible deployment, the system's exact boundaries and limitations must be clearly understood:

- **Does NOT Read Thoughts or Brainwaves:** It evaluates outward physiological and biometric indicators (eyes, face angle, voice), not neurological thoughts or internal intent.
- **NOT a Legally Admissible Medical Polygraph:** This software is designed for simulation, research, educational demonstrations, and entertainment. It cannot replace forensic polygraph examinations in a court of law.
- **Does NOT Operate in Complete Darkness:** Optical computer vision requires minimum ambient facial lighting (>20 lux) to extract eye socket landmarks.
- **Does NOT Work When Masked or Obstructed:** If sunglasses, thick face masks, or heavy obstructions cover the eyes, eye contact cannot be computed.
- **Does NOT Force 100% Constant Lies or Constant Truth:** The algorithm is balanced. If the subject maintains camera eye contact, it yields TRUTH. If eye contact breaks, it yields LIE.

3. How It Works (Step-by-Step Architecture)

The application is structured into an interconnected three-tier pipeline:

Pipeline Layer	Component & Technology	Operational Function
Hardware & Vision	OpenCV (cv2) + Haar Cascades	Captures camera 0/1, detects face bounding box, isolates upper 55% eye region, and tracks eye pupils.
Telemetry API	Flask REST + MJPEG Stream	Streams annotated video via /video_feed and exposes live telemetry via /api/biometrics.
Acoustic Engine	Web Audio API + Autocorrelation	Captures microphone PCM stream, estimates fundamental voice pitch, and calculates pitch jitter/tremor percentage.
Polygraph HUD	HTML5 Canvas + Cyberpunk CSS	Renders ECG cardio wave, voice waveform, eye tension stylus, live stress gauge, and target reticles.
Decision Fusion	LieDetectorEngine (JS)	Fuses eye contact ratio (40%), vocal jitter (35%), and blink dynamics (25%) over 5-second interrogation.

The Core Eye-Contact Decision Formula:

- **Direct Eye Contact (Eye Contact ≥ 50%):** When the subject looks directly into the webcam lens, both eyes remain consistently detected in the upper facial ROI. The system tags the frame with a **GREEN BOX** [DIRECT EYE CONTACT: TRUTH] and finalizes a **TRUTH VERIFIED** verdict.
- **Averted Gaze / Looking Away (Eye Contact < 50%):** When the subject shifts their gaze away, looks down, or turns their head, eye detection count drops to 0 or 1. The bounding box switches to **RED** [EYE CONTACT BROKEN: LIE] and triggers a **LIE DETECTED** verdict.

4. How It Does NOT Work (Failure Modes & Edge Cases)

Awareness of specific failure modes ensures correct operation and diagnostic troubleshooting:

- **Hardware Port Contention:** If another application (Zoom, Microsoft Teams, OBS Virtual Camera) holds an exclusive lock on Camera 0, OpenCV will report error -1072875772. The user must release camera access in other programs or use the SWITCH CAM button.

• **Audio Context Mute:** If the browser's audio context is in a suspended state without user interaction, sound effects will be muted until the user clicks anywhere on the dashboard.

• **Multiple Subjects in View:** The vision pipeline isolates the largest detected face. If multiple persons enter the camera frame simultaneously, background movements may interfere with gaze scoring.

• **No User Answer Action:** The polygraph does not guess arbitrarily; the user must speak their answer during the 5-second active scan (triggered by SPACEBAR or the START LIE TEST button).

5. The Monitor HUD Dashboard (Layout & Indicators)

The user interface operates as an authentic military/cyberpunk polygraph terminal divided into two primary zones:

• **Channel 01 - Optical Biometric Viewport:** Displays the 30 FPS mirrored camera feed with dynamic green/red reticles, pupil target crosshairs, and live status badges (FACIAL LOCK ACTIVE, GAZE: DIRECT CONTACT).

• **Biometric Metric Cards:** Three digital telemetry readouts displaying **Total Blinks**, **Blink Frequency (BPM)**, **Eye Contact Stability (%)**, and **Eye Aperture (EAR)**.

• **Channel 02 - Polygraph Oscillograph:** Continuous multi-channel scrolling wave canvas showing real-time spikes for Cardio ECG pulse, vocal frequency tension, ocular jitter, and deception probability.

• **Channel 03 - Vocal Tremor & Spectrum:** 36-band audio equalizer canvas reflecting live voice frequencies, pitch jitter percentage, and vocal micro-tremor classification (STEADY, ELEVATED, HIGH TREMOR).

• **Suspicion / Deception Meter:** Horizontal gradient gauge (0 to 100%) with a digital numeric readout and a clearly marked 48% Lie Threshold marker.

• **Interrogation Control Unit:** Displays English interrogation prompt bank with PREV and NEXT buttons, plus the primary **START LIE TEST** trigger button.

• **The Verdict Modal:** Slams into the viewport with emergency red sirens (LIE) or toxic green holographic shields (TRUTH) with four forensic evidence breakdown cards.

6. Technology Stack & Tools Used

Category	Library / Tool	Version / Source	Role in Project
Programming Language	Python	v3.12.10	Core server, hardware vision loop, API controller.
Web Framework	Flask	v3.1.3	Serves HTTP frontend, MJPEG stream, and JSON telemetry.
Computer Vision	OpenCV (cv2)	v5.0.0.93	Hardware video capture, Haar cascades, facial drawing.
Vision Models	Haar Face & Eye XMLs	OpenCV Official	Frontal face and dual eye bounding box extraction.
Frontend Stack	HTML5 / CSS3 / ES6+	Native Standards	Cyberpunk CRT HUD, scanline animations, responsive grid.
Acoustic Processing	Web Audio API	W3C Standard	Microphone pitch tracking, spectrum FFT, audio oscillators.
Sound Synthesis	Procedural Synth	Custom JS	Zero-file audio: Heartbeat, scan tension, siren, chime.
Report Generator	ReportLab	v5.1.0	Automated PDF forensic technical report generation.